

Phase I/II Operating Characteristics

TITE versus new non-TITE: one-stage, two-stage highest utility, and two-stage top-2 randomized

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 1						
DLT rate	0.3	0.35	0.4	0.45	0.5	
Efficacy rate	0.05	0.06	0.07	0.08	0.09	
OBD selection (%)						
TITE one-stage	18.2	9.1	1.8	0.4	0	70.5
New non-TITE one-stage	17.4	6.4	1.8	0.1	0.2	74.1
TITE 2-stage highest utility	18.6	8.8	0.9	0.1	0	71.6
New non-TITE 2-stage highest utility	18	7.2	1.2	0	0.2	73.4
TITE 2-stage top-2 randomized	18.6	8.8	0.9	0.1	0	71.6
New non-TITE 2-stage top-2 randomized	17.4	7.2	2	0.8	0.1	72.5
Mean administrations by dose						
TITE one-stage	11.5	4.7	2.1	0.9	0.3	
New non-TITE one-stage	11.1	4.5	2.2	0.9	0.3	
TITE 2-stage highest utility	11.5	4.6	2.2	0.9	0.3	
New non-TITE 2-stage highest utility	11.1	4.4	2.3	1	0.3	
TITE 2-stage top-2 randomized	11.5	4.6	2.2	0.9	0.3	
New non-TITE 2-stage top-2 randomized	11	4.4	2.3	1	0.3	
Scenario 16						
DLT rate	0.02	0.06	0.09	0.14	0.26	
Efficacy rate	0.2	0.2	0.2	0.2	0.2	
OBD selection (%)						
TITE one-stage	29.2	22.6	19.8	15.5	12.8	0.1
New non-TITE one-stage	28.7	21.3	20.9	18.1	11	0
TITE 2-stage highest utility	31.4	24.3	19.3	15.1	9.8	0.1
New non-TITE 2-stage highest utility	34.1	21.3	16.6	16.4	11.5	0.1
TITE 2-stage top-2 randomized	31.4	24.3	19.3	15.1	9.8	0.1
New non-TITE 2-stage top-2 randomized	33.9	22.2	16.5	15.4	12	0
Mean administrations by dose						
TITE one-stage	7.1	6.8	5.9	5.7	4.6	
New non-TITE one-stage	7.3	6.6	5.7	5.7	4.7	
TITE 2-stage highest utility	7.3	7	5.9	5.5	4.3	
New non-TITE 2-stage highest utility	7.6	6.8	5.7	5.5	4.4	
TITE 2-stage top-2 randomized	7.3	7	5.9	5.5	4.3	
New non-TITE 2-stage top-2 randomized	6.6	6.1	5.6	6.1	5.6	

Sources: TITE IDX 1001-2000 and new non-TITE IDX 0001-1000, both under Presentation 8-17. All populated rows represent 1,000 simulations.

All three designs are present in both sources. TITE No OBD % = 100 - sum of dose OBD selections.

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Scenario 20						
DLT rate	0.07	0.14	0.31	0.41	0.52	
Efficacy rate	0.05	0.25	0.25	0.25	0.25	
OBD selection (%)						
TITE one-stage	10.1	45.3	26.7	13.5	1.6	2.8
New non-TITE one-stage	15.2	44	25.1	11.4	1.6	2.7
TITE 2-stage highest utility	11	50.2	25.9	8.8	1.2	2.9
New non-TITE 2-stage highest utility	14.2	47.4	25.8	7.7	1.7	3.2
TITE 2-stage top-2 randomized	11	50.2	25.9	8.8	1.2	2.9
New non-TITE 2-stage top-2 randomized	10.4	47.7	25.6	10.6	2.7	3
Mean administrations by dose						
TITE one-stage	5.1	10.6	7.9	4.5	1.4	
New non-TITE one-stage	5.2	11.1	7.6	4.2	1.4	
TITE 2-stage highest utility	5.4	11.1	8	3.8	1.4	
New non-TITE 2-stage highest utility	5.3	11.2	7.9	3.8	1.3	
TITE 2-stage top-2 randomized	5.4	11.1	8	3.8	1.4	
New non-TITE 2-stage top-2 randomized	5.7	9.5	8	4.6	1.8	
Scenario 24						
DLT rate	0.01	0.02	0.05	0.1	0.25	
Efficacy rate	0.05	0.08	0.2	0.5	0.5	
OBD selection (%)						
TITE one-stage	1.1	1.1	4.2	59.7	33.9	0
New non-TITE one-stage	1.4	1.4	4.7	59.1	33.4	0
TITE 2-stage highest utility	1.3	1.8	4.8	56	36.1	0
New non-TITE 2-stage highest utility	2	1.5	6.2	55.5	34.8	0
TITE 2-stage top-2 randomized	1.3	1.8	4.8	56	36.1	0
New non-TITE 2-stage top-2 randomized	0.4	0.8	3.6	56.6	38.6	0
Mean administrations by dose						
TITE one-stage	3.4	3.4	4.4	12.2	6.5	
New non-TITE one-stage	3.4	3.5	4.4	12.3	6.4	
TITE 2-stage highest utility	3.4	3.5	4.5	10.6	8	
New non-TITE 2-stage highest utility	3.4	3.5	4.4	10.6	8.1	
TITE 2-stage top-2 randomized	3.4	3.5	4.5	10.6	8	
New non-TITE 2-stage top-2 randomized	3.6	3.7	4.9	9.4	8.4	

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	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 27						
DLT rate	0.11	0.17	0.3	0.5	0.62	
Efficacy rate	0.1	0.2	0.4	0.42	0.45	
OBD selection (%)						
TITE one-stage	13	26.5	45.5	9.3	0.6	5.1
New non-TITE one-stage	16.3	21.2	44.7	11.3	0.7	5.8
TITE 2-stage highest utility	16.4	27.8	42.4	6.8	0.5	6.1
New non-TITE 2-stage highest utility	16.3	23.9	44	8.4	0.9	6.5
TITE 2-stage top-2 randomized	16.4	27.8	42.4	6.8	0.5	6.1
New non-TITE 2-stage top-2 randomized	10.5	23.6	46.6	12.5	1.1	5.7
Mean administrations by dose						
TITE one-stage	6.8	8	9.8	3.9	0.6	
New non-TITE one-stage	7.1	7.9	10	3.6	0.7	
TITE 2-stage highest utility	7.7	8.2	9	3.3	0.9	
New non-TITE 2-stage highest utility	7.7	7.9	9.4	3.2	0.9	
TITE 2-stage top-2 randomized	7.7	8.2	9	3.3	0.9	
New non-TITE 2-stage top-2 randomized	7.4	8.2	8.4	4	1.1	
Scenario 38						
DLT rate	0.05	0.1	0.12	0.15	0.3	
Efficacy rate	0.05	0.08	0.12	0.15	0.5	
OBD selection (%)						
TITE one-stage	13.9	10.3	7.8	8.2	58.5	1.3
New non-TITE one-stage	13.8	7.2	6.8	7.8	63.5	0.9
TITE 2-stage highest utility	17.2	10	8.9	8.9	53.5	1.5
New non-TITE 2-stage highest utility	17.9	10	7.3	7.5	56	1.3
TITE 2-stage top-2 randomized	17.2	10	8.9	8.9	53.5	1.5
New non-TITE 2-stage top-2 randomized	11.7	6	5.5	8.5	67.3	1
Mean administrations by dose						
TITE one-stage	4.6	4.8	5	5.6	9.8	
New non-TITE one-stage	4.6	4.6	5	5.3	10.4	
TITE 2-stage highest utility	4.9	5	5.5	5.4	9	
New non-TITE 2-stage highest utility	4.8	4.7	5.3	5.4	9.6	
TITE 2-stage top-2 randomized	4.9	5	5.5	5.4	9	
New non-TITE 2-stage top-2 randomized	4.7	4.8	5.3	6.9	8.2	